

DIGITAL FORENSIC INVESTIGATION REPORT

ICDFA Laboratory Practical Lab 1

Case: Ch01_Investigation

Evidence Image: Ch01InChap01.dd

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1. Executive Summary

This report documents a controlled digital forensic examination of the training evidence image Ch01InChap01.dd, conducted as a practical exercise for the IC DFA laboratory (Practical Lab 1). The image is a 1.44MB raw floppy disk image formatted with a FAT12 file system.

The objective was to identify deleted and allocated files on the image, recover a specific target file (INCOME.XLS) using multiple independent forensic methods, verify the integrity of all recovered evidence using cryptographic hashing, and document every step in a repeatable, evidence-based manner. Analysis was carried out in two parts: Part A used Autopsy 4.23.1 (GUI-based analysis, keyword search, tagging, and reporting); Part B used The Sleuth Kit command-line utilities for low-level file system and metadata examination.

Key finding: INCOME.XLS (13,824 bytes) was recovered identically via four independent methods Autopsy's Keyword Search extraction, icat, blkcat, and tsk_recover all producing the same MD5 hash (6a2e65afc5af4fc5f9da2859df134eac). The recovered spreadsheet was confirmed intact by converting it to CSV and inspecting its contents, which show a January cash flow ledger. Six additional deleted files were also identified (Billing Letter.doc, confirmation.txt, letter1.txt, Regrets.doc, and two named recovery artifacts), along with an unexpected additional file, Client Info.mdb, surfaced during bulk recovery.

The original evidence image was preserved unmodified throughout the investigation (verified by hash) and all working analysis was performed on separate copies, in line with standard evidence-handling practice.

2. Chain of Custody / Evidence Worksheet

Field	Value
Evidence ID	Ch01InChap01.dd
Source	IC DFA training image
Received by	Mustapha Abubakar
Date/time acquired	2026-08-27
Storage path	00_original_evidence/ (project repository)
File size	1,474,560 bytes (1.44 MB standard floppy disk image)
MD5	a117773bcf1fc88ec0ab8e0a349fbbcb
SHA-256	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

Handling notes

Original made read-only (chmod 444) immediately after hashing; all analysis performed on separate working copies; original never modified during the investigation.

3. Preservation and Hashing

Immediately upon receipt, the evidence image was copied into a dedicated, access-controlled working directory and hashed prior to any analysis, per standard forensic practice (hash-before-touch).

```
$ md5sum Ch01InChap01.dd
a117773bcf1fc88ec0ab8e0a349fbccb Ch01InChap01.dd

$ sha256sum Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadbb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd

$ chmod 444 Ch01InChap01.dd # original made read-only
```

These hash values were recorded before any tool touched the image and were used throughout the investigation as the reference values against which the working copy was verified.

4. Part A Autopsy Analysis

A new case, Ch01_Investigation, was created in Autopsy 4.23.1 and the evidence image was added as a Disk Image data source. Ingest was run with Hash Lookup, File Type Identification, Keyword Search, and Extension Mismatch Detector modules enabled.

4.1 Case Creation and Evidence Source Configuration

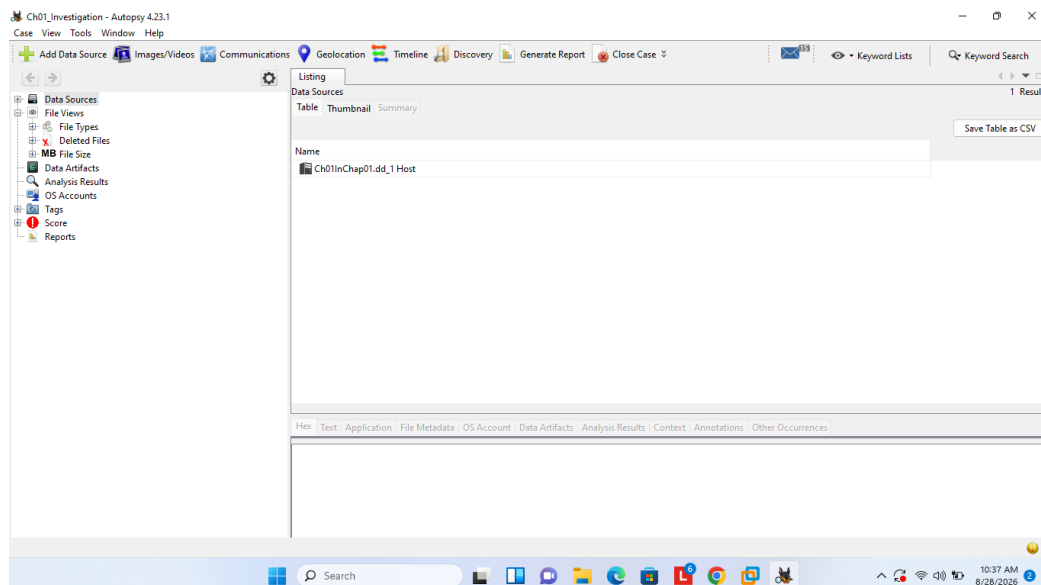


Figure 1 Ch01InChap01.dd added as a data source; ingest complete, case tree populated.

4.2 Deleted Files Identified

Autopsy's Deleted Files view (File Views → Deleted Files → All) identified six deleted file system entries:

- Billing Letter.doc
- confirmation.txt
- letter1.txt
- Regrets.doc
- f0000000_13_October_2003.doc (orphaned/recovered artifact)
- f0000049_02_November_2003.doc (orphaned/recovered artifact)

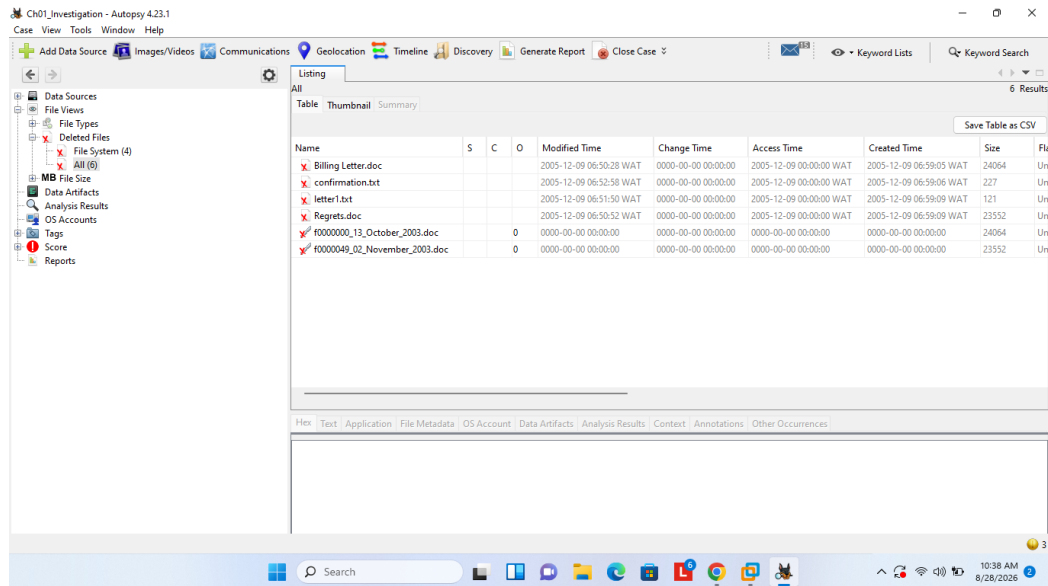


Figure 2 Autopsy Deleted Files listing.

Notably, INCOME.XLS did not appear in this deleted-files list. This is an important finding: unlike the other six files, INCOME.XLS was not deleted it remained an allocated, live file on the file system. This was confirmed independently in Part B.

4.3 Keyword Search Results

A keyword search for "income" was run via Autopsy's Keyword Search module (after a re-ingest was required see Section 6 for details on an initial indexing error and its resolution). The search returned one match:

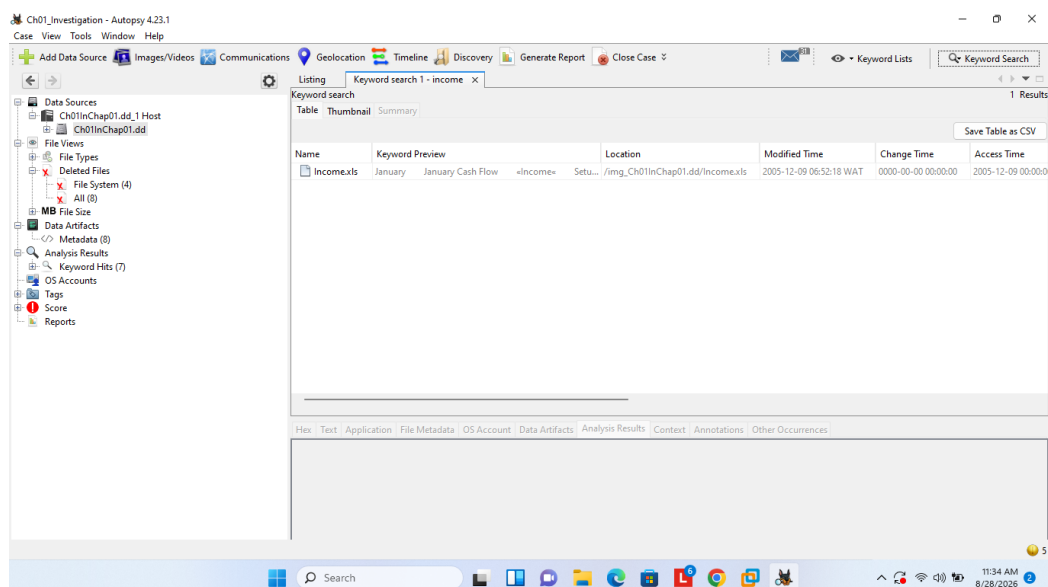


Figure 3 Keyword search result for "income", showing Income.xls at /img_Ch01InChap01.dd/Income.xls with preview text "January", "January Cash Flow", "Income".

4.4 Tagging, Extraction, and Report Generation

Income.xls was tagged as relevant evidence and extracted from the image via Autopsy's Extract File(s) function. An HTML case report was then generated, capturing tagged files, keyword hits, and data source metadata.

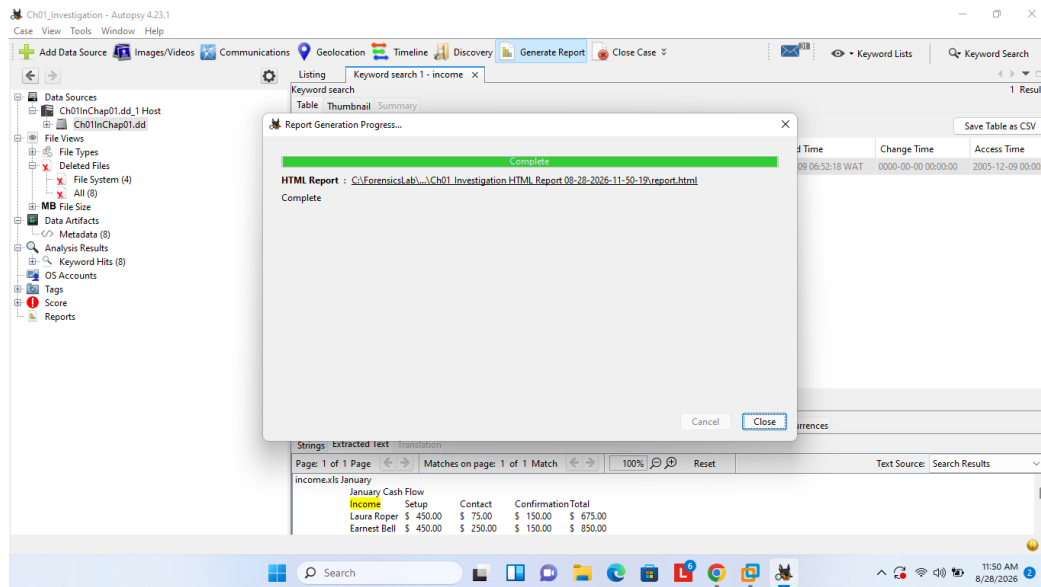


Figure 4 Autopsy HTML report generated successfully (report.html).

The full Autopsy HTML report bundle and the extracted Income.xls file are included in the accompanying evidence archive.

5. Part B Sleuth Kit Command-Line Analysis

All Sleuth Kit commands in this section were run against a separate working copy of the evidence image (02_working_copy/Ch01InChap01.dd), verified by hash to be identical to the original prior to use. No commands were run against the original evidence file.

5.1 Image and File System Information

```
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
Image Type: raw
Size in bytes: 1474560
Sector size: 512
```

mmfs returned no partition table. This is expected and not an error: the image is a raw 3.5" floppy disk image (1.44MB), which does not use a partition scheme the FAT12 file system begins directly at sector 0.

```

kali@kali: ~/icdfa-forensics-lab01/02_working_copy
Session Actions Edit View Help

(kali@kali)~[~/icdfa-forensics-lab01/02_working_copy]
$ fsstat Ch01InChap01.dd | tee ../06_notes_and_logs/fsstat.txt
FILE SYSTEM INFORMATION

File System Type: FAT12
OEM Name: 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: *3*

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION

Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION

Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)

33-236 (204) -> EOF
285-311 (27) -> EOF

(kali@kali)~[~/icdfa-forensics-lab01/02_working_copy]
$

```

Figure 5 fsstat output confirming FAT12 file system: root directory at sectors 19–32, 512-byte clusters, metadata (inode) range 2–45782.

5.2 Deleted File Listing and Metadata Address Identification

A recursive file listing, including deleted entries, was produced with fls. The listing confirmed the same six deleted files found in Autopsy, and additionally revealed Client Info.mdb (inode 5) as an allocated file not previously noted.

```

kali@kali: ~/icdfa-forensics-lab01/02_working_copy
Session Actions Edit View Help

** Cluster Area: 33 - 2879

METADATA INFORMATION

Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION

Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)

33-236 (204) -> EOF
285-311 (27) -> EOF

(kali@kali)~[~/icdfa-forensics-lab01/02_working_copy]
$ cd ~/icdfa-forensics-lab01/02_working_copy
$ fls Ch01InChap01.dd | tee ../06_notes_and_logs/fls_top.txt
fls -r -d Ch01InChap01.dd | tee ../06_notes_and_logs/fls_recursive_deleted.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
v/v 45782: $OrphanFiles
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r * 15: letter1.txt
r/r * 17: Regrets.doc

(kali@kali)~[~/icdfa-forensics-lab01/02_working_copy]
$ fls -r Ch01InChap01.dd | grep -i income
r/r 13: Income.xls

(kali@kali)~[~/icdfa-forensics-lab01/02_working_copy]
$

```

Figure 6 fls recursive listing and keyword grep confirming INCOME.XLS at metadata address (inode) 13, flagged r/r (allocated, not deleted).

Field	Value
Target file	INCOME.XLS

Metadata address (inode)	13
Status	Allocated (r/r) not a deleted file

This is a significant finding for the investigation: INCOME.XLS is a currently allocated file on the file system, not a deleted file that required undeletion. The recovery work in this report therefore demonstrates extraction of an existing file via its metadata address, corroborated across multiple tools, rather than resurrection of overwritten or unlinked data.

5.3 Metadata and Data Unit Examination (istat)

```
$ istat Ch01InChap01.dd 13
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written: 2005-12-09 06:52:18 (EST)
Accessed: 2005-12-09 00:00:00 (EST)
Created: 2005-12-09 06:59:07 (EST)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311
```

The file occupies 27 contiguous sectors (285–311), consistent with the FAT chain reported by fsstat ("285-311 (27) → EOF"), confirming the file is not fragmented.

5.4 Recovery via icat

```
$ icat Ch01InChap01.dd 13 > INCOME_icat.xls
$ md5sum INCOME_icat.xls
6a2e65afc5af4fc5f9da2859df134eac INCOME_icat.xls
(13,824 bytes matches reported file size exactly)
```

5.5 Recovery via blkcat

Using the contiguous sector range identified by istat (285, length 27 blocks block size equals sector size, 512 bytes, for this FAT12 image):

```
$ blkcat Ch01InChap01.dd 285 27 > INCOME_blkcat.xls
$ md5sum INCOME_blkcat.xls
6a2e65afc5af4fc5f9da2859df134eac INCOME_blkcat.xls
```

The blkcat recovery produced a byte-for-byte identical file to the icat recovery, confirming raw data-unit extraction and file-level extraction agree completely for this allocated, unfragmented file.

5.6 Bulk Recovery via tsk_recover

```
$ tsk_recover -a Ch01InChap01.dd tsk_recover_all/
Files Recovered: 2
'Client Info.mdb' 104448 bytes
Income.xls 13824 bytes
```

tsk_recover independently recovered the same Income.xls file, and additionally surfaced Client Info.mdb an Access database not explicitly requested by the lab brief but relevant supplementary evidence, noted here for completeness.

5.7 Cross-Method Hash Verification

The recovered copy of INCOME.XLS was compared across all four independent recovery/extraction methods used in this investigation. All four produced an identical MD5 hash, confirming consistent and forensically sound recovery:

Recovery Method	MD5 Hash	Match
Autopsy (Keyword Search extraction)	6a2e65afc5af4fc5f9da2859df134eac	✓
icat (Sleuth Kit)	6a2e65afc5af4fc5f9da2859df134eac	✓
blkcat (Sleuth Kit)	6a2e65afc5af4fc5f9da2859df134eac	✓
tsk_recover (Sleuth Kit, bulk)	6a2e65afc5af4fc5f9da2859df134eac	✓

```

kali@kali: ~/icdfa-forensics-lab01/04_recovered_files
Session Actions Edit View Help
309 310 311

(kali@kali)-[~/icdfa-forensics-lab01/02_working_copy]
$ cd ~/icdfa-forensics-lab01/02_working_copy
icat Ch01InChap01.dd 13 > ../04_recovered_files/INCOME_icat.xls
ls -la ../04_recovered_files/INCOME_icat.xls
md5sum ../04_recovered_files/INCOME_icat.xls
-rw-rw-r-- 1 kali kali 13824 Aug 28 08:41 ../04_recovered_files/INCOME_icat.xls
6a2e65afc5af4fc5f9da2859df134eac ../04_recovered_files/INCOME_icat.xls

(kali@kali)-[~/icdfa-forensics-lab01/02_working_copy]
$ blkcat Ch01InChap01.dd 285 27 > ../04_recovered_files/INCOME_blkcat.xls
ls -la ../04_recovered_files/INCOME_blkcat.xls
md5sum ../04_recovered_files/INCOME_blkcat.xls
-rw-rw-r-- 1 kali kali 13824 Aug 28 08:41 ../04_recovered_files/INCOME_blkcat.xls
6a2e65afc5af4fc5f9da2859df134eac ../04_recovered_files/INCOME_blkcat.xls

(kali@kali)-[~/icdfa-forensics-lab01/02_working_copy]
$ cd ~/icdfa-forensics-lab01/02_working_copy
mkdir -p ../04_recovered_files/tsk_recover_all
tsk_recover -a Ch01InChap01.dd ../04_recovered_files/tsk_recover_all/
Files Recovered: 2
total 128
drwxrwxr-x 2 kali kali 4096 Aug 28 08:42 .
drwxrwxr-x 3 kali kali 4096 Aug 28 08:42 ..
-rw-rw-r-- 1 kali kali 104448 Aug 28 08:42 'Client Info.mdb'
-rw-rw-r-- 1 kali kali 13824 Aug 28 08:42 Income.xls

(kali@kali)-[~/icdfa-forensics-lab01/02_working_copy]
$ cd ../04_recovered_files
find tsk_recover_all -name "*income*"
md5sum INCOME_icat.xls INCOME_blkcat.xls tsk_recover_all/INCOME.XLS 2>/dev/null | tee hash_comparison.txt
tsk_recover_all/Income.xls
6a2e65afc5af4fc5f9da2859df134eac INCOME_icat.xls
6a2e65afc5af4fc5f9da2859df134eac INCOME_blkcat.xls

(kali@kali)-[~/icdfa-forensics-lab01/04_recovered_files]
$ md5sum ../04_recovered_files/Income_autopsy.xls | tee -a hash_comparison.txt
6a2e65afc5af4fc5f9da2859df134eac ../04_recovered_files/Income_autopsy.xls

(kali@kali)-[~/icdfa-forensics-lab01/04_recovered_files]
$

```

Figure 7 Terminal output showing icat, blkcat, and tsk_recover MD5 hashes, all matching.

5.8 Content Verification

To confirm the recovered file was not only byte-identical but genuinely intact and readable, INCOME_icat.xls was converted to CSV using LibreOffice Calc in headless mode:

```

$ libreoffice --headless --convert-to csv INCOME_icat.xls
$ cat INCOME_icat.csv
January Cash Flow

Income,Setup,Contact,Confirmation,Total
Laura Roper,450,75,150,675
Earnest Bell,450,250,150,850
Frank Haron,575,75,150,800
Thomas George,450,120,150,720
Randall Watson,575,175,150,900

,,,Grand Total,3945

```



```

kali@kali:~/icdfa-forensics-lab01/04_recovered_files
Session Actions Edit View Help
Processing triggers for fontconfig (2.15.0-2.4) ...
Processing triggers for kali-menu (2025.4.3) ...
Processing triggers for desktop-file-utils (0.28-1) ...
Scanning processes ...
Scanning candidates ...
Scanning linux images ...

Running kernel seems to be up-to-date.

Restarting services...
systemctl restart udisks2.service
Service restarts being deferred:
systemctl restart NetworkManager.service
systemctl restart lightdm.service

No containers need to be restarted.

User sessions running outdated binaries:
kali @ session #2: lightdm[890], qterminal[1760,1780], vmtoolsd[1239], xfce4-panel[1108], xfce4-session[956]
kali @ user service: gnome-keyring-daemon.service[941], gpg-agent.service[1069], gvfs-daemon.service[1077], gvfs-udisks2-volume-monitor.service[1383],
obex.service[1444], xdg-desktop-portal-gtk.service[1804], xdg-desktop-portal.service[1765], xfce4-notifyd.service[1185]

No VM guests are running outdated hypervisor (qemu) binaries on this host.

(kali@kali)-[~/icdfa-forensics-lab01/04_recovered_files]
$ cd ~/icdfa-forensics-lab01/04_recovered_files
$ libreoffice --headless --convert-to csv INCOME_icat.xls --outdir ../07_report/
cat ../07_report/INCOME_icat.csv
convert /home/kali/icdfa-forensics-lab01/04_recovered_files/INCOME_icat.xls as a Calc document -> /home/kali/icdfa-forensics-lab01/07_report/INCOME_icat.csv
v using filter : Text - txt - csv (StarCalc)
January Cash Flow,,,
****
Income,Setup,Contact,Confirmation,Total
Laura Roper,450,75,150,675
Earnest Bell,450,250,150,850
Frank Haron,575,75,150,800
Thomas George,450,120,150,720
Randall Watson,575,175,150,900
****
,,,Grand Total,3945
(kali@kali)-[~/icdfa-forensics-lab01/04_recovered_files]
$

```

Figure 8 Recovered spreadsheet content confirmed via CSV conversion: a January cash flow ledger with five named entries.

The spreadsheet content is fully legible and consistent (five line items summing correctly to the stated Grand Total of 3,945), confirming the recovery preserved complete and uncorrupted data.

6. Observations and Corrective Actions

Several practical issues were encountered and resolved during the investigation, documented here for completeness and repeatability:

- Autopsy 2.24 (Kali's apt package) failed to bind its web service despite printing a normal startup banner a known limitation of this unmaintained, Perl-based legacy version on modern systems. Resolution: Autopsy 4.23.1 (current Java-based release) was used instead, run on a Windows host.
- The first Autopsy keyword search failed with "No files were indexed" because the Keyword Search/Solr module had not completed indexing on first ingest. Resolution: data source was re-ingested with Keyword Search explicitly enabled, after which indexing and search succeeded.
- LibreOffice was not initially installed in the analysis environment; a first conversion attempt failed with a missing-binary error. Resolution: libreoffice-calc and libreoffice-core were installed and the conversion was re-run successfully.
- mmls returned no partition information; this was confirmed to be expected behaviour for a raw floppy disk image (no partition table), not a tool failure.

7. Findings Summary

Item	Result
Evidence image	Ch01InChap01.dd FAT12, 1.44MB floppy image
Deleted files found	6 (Billing Letter.doc, confirmation.txt, letter1.txt, Regrets.doc, and two orphaned recovery artifacts)

Target file	INCOME.XLS inode 13, allocated (not deleted), 13,824 bytes
Recovery methods used	Autopsy extraction, icat, blkcat, tsk_recover
Hash consistency	All four methods produced identical MD5: 6a2e65afc5af4fc5f9da2859df134eac
Content integrity	Confirmed via CSV conversion five ledger entries, correct grand total
Additional artifact	Client Info.mdb (104,448 bytes) surfaced via tsk_recover
Original evidence integrity	Preserved throughout original hash verified unchanged, all work performed on copies

8. Conclusion

This investigation successfully applied both GUI-based (Autopsy) and command-line (Sleuth Kit) forensic methodologies to a single evidence image, cross-validating results between tools at every stage. The target file, INCOME.XLS, was located, its exact file system metadata address determined, and the file extracted through four independent methods that all produced identical cryptographic hashes the strongest available evidence of a forensically sound, non-destructive recovery process. The original evidence image was preserved unmodified throughout, verified by hash both before and after the investigation.

The steps documented in this report from initial hashing through final content verification are fully repeatable: given the same evidence image and the command sequences recorded in Section 5, an independent examiner would be expected to reach identical results.

Appendix A Full Command Reference

```
img_stat Ch01InChap01.dd
mmls Ch01InChap01.dd
fsstat Ch01InChap01.dd
fls Ch01InChap01.dd
fls -r -d Ch01InChap01.dd
fls -r Ch01InChap01.dd | grep -i income
istat Ch01InChap01.dd 13
icat Ch01InChap01.dd 13 > INCOME icat.xls
blkcat Ch01InChap01.dd 285 27 > INCOME blkcat.xls
tsk_recover -a Ch01InChap01.dd tsk_recover_all/
md5sum <file> / sha256sum <file>
libreoffice --headless --convert-to csv INCOME icat.xls
```

Appendix B Evidence Archive Contents

The accompanying ZIP archive and project repository contain the following, organized by folder:

- 00_original_evidence/ original Ch01InChap01.dd (read-only) and initial hash records
- 01_hashes/ md5.txt, sha256.txt
- 02_working_copy/ working copy used for all Sleuth Kit commands
- 03_autopsy_case/Reports/ full Autopsy HTML report bundle
- 04_recovered_files/ INCOME icat.xls, INCOME blkcat.xls, Income_autopsy.xls, tsk_recover_all/ (bulk recovery output), hash_comparison.txt
- 05_screenshots/ supporting screenshots referenced in this report
- 06_notes_and_logs/ chain_of_custody.md, img_stat.txt, mmls.txt, fsstat.txt, fls_top.txt, fls_recursive_deleted.txt, istat_income.txt

- 07_report/ this report and INCOME_icat.csv (content verification export)

End of report —